

PROTECTING AGAINST ESD

From The Floor Up



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Electrostatic discharge (ESD) can pose a serious threat to sensitive electronic equipment. It can damage circuitry, disable components and generally pose a threat to the working life of electrical equipment. In the long run, if a business in the electronics industry is unable to clamp down on this threat, ESDs can have a significant impact on productivity, product reliability, rework costs and ultimately profitability. In fact, ESD Association estimates that as much as 25 per cent of all electronics that get damaged for unknown reasons can be attributed to ESD, and that in total ESDs cost the electronics industry US\$ 5 billion a year.

An ESD event can permanently damage a device's circuitry so that it stops functioning entirely or partially by causing a variety of problems such as metal melt, junction breakdown or oxide failure. If the damage happens after the item's final test in a plant, the product could be sold or used despite this fault. When the flaw leads to a problem further down the line, it can have repercussions on reputation of the manufacturer.

Specialist conductive floor finish at Toshiba Medical Systems, Malaysia



ESD causes

An ESD event could take place at any point in the product's lifecycle, from its production to being put into operation, and normally it results from a device being handled in an uncontrolled environment or with poor ESD-control practices. This makes it imperative for electronics manufacturers to ensure that they have effective static-control measures in place; otherwise they will struggle to routinely produce high-quality devices.

Certain materials coming into contact with each other and then separating generate an electrostatic charge. This can be caused by a long list of things such as when a device slides into a bag. However, one of the most common causes is people walking across the floor; with each step the contact and separation of the sole of the shoe with the floor's surface creates a small charge.

This charge can quickly build up when it is subjected to regular traffic from people and wheeled equipment. Factors such as the material of the floor, type of footwear and even humidity can have an effect on the amount of charge generated.

One of the most common materials that generate ESD is human skin itself, as it is very conductive in electrostatic terms. If a person is electrically-isolated by synthetic soles of shoes, the charge can accumulate on the body like in a capacitor, with voltage potentially rising to several kilovolts.

You are unlikely to know that you are carrying around this charge until you touch a grounded object and receive a shock, as the body releases all of its accumulated charge at once and a spark jumps from the body to the object. If it is an electrical device that you touched, it could inadvertently get damaged.

Antistatic flooring

Antistatic flooring solutions can reduce this risk. However, there are several things to